

$\therefore \exists n \forall p (p \neq (m+n)/2)$ is False. for all
 By defn. of existential quantifier, ~~there exists an~~ integer a s.t.
 $\exists m \forall p (p \neq (a+n)/2)$ is False. By defn. of existential quantifier,
 for all integers b in the domain $\forall p (p \neq (a+b)/2)$ is False.

By defn. of universal quantifier, ~~there exists an~~ there exists an
 integer c in the domain s.t. $(c \neq (a+b)/2)$ is False.

$\therefore$ For all integers a and for all integers b , there exists an integer
 c s.t. $(c \neq (a+b)/2)$ is False, or $(c = (a+b)/2)$ is True.

Let $a=3, b=2$ $c = (3+2)/2 = 2.5$ ~~2.5~~ which is not an Integer.

$\therefore$ This is a contradiction

$\therefore \forall n \forall m \exists p (p = (m+n)/2)$ is False. $\square$

29) Suppose the domain of the propositional function $P(x,y)$ consists
 of pairs x and y , where x is 1, 2 or 3 and y is 1, 2 or 3. Write out
 these propositions using disjunctions and conjunctions.

a) $\forall x \forall y P(x,y) : (P(1,1) \wedge P(1,2) \wedge P(1,3)) \wedge (P(2,1) \wedge P(2,2) \wedge P(2,3)) \wedge (P(3,1) \wedge P(3,2) \wedge P(3,3))$

b) $\exists x \exists y P(x,y) : (P(1,1) \vee P(1,2) \vee P(1,3)) \vee (P(2,1) \vee P(2,2) \vee P(2,3)) \vee (P(3,1) \vee P(3,2) \vee P(3,3))$

c) $\exists x \forall y P(x,y) : (P(1,1) \wedge P(1,2) \wedge P(1,3)) \vee (P(2,1) \wedge P(2,2) \wedge P(2,3)) \vee (P(3,1) \wedge P(3,2) \wedge P(3,3))$

d) $\forall y \exists x P(x,y) : (P(1,1) \vee P(2,1) \vee P(3,1)) \wedge (P(1,2) \vee P(2,2) \vee P(3,2)) \wedge (P(1,3) \vee P(2,3) \vee P(3,3))$

30) Rewrite each of these statements so that negations appear only within
 predicates (that is, so that no negation is outside a quantifier or an
 expression involving logical connectives).

a) $\neg \exists y \exists x P(x,y) \equiv \forall y \neg \exists x P(x,y)$ (By De Morgan's Law of quantifiers)
 $\equiv \forall y \forall x \neg P(x,y)$ (By De Morgan's Law of quantifiers)